# New Features in Wireless InSite 3.4.4

**ELECTROMAGNETICALLY ENGINEERED SURFAFCES**

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7. Numerous fixes and enhancements to resolve bugs, improve usability and enhance performance

## Electromagnetically Engineered Surfafces

### 1. CRC Ray-Optical Scattering Model

Remcom has integrated Communications Research Centre Canada's (CRC) ray-optical Engineered Electromagnetic Surfaces (EESs) scattering model (Ray-Optical or "RO" Model) into the X3D Model. This enables the user to simulate EM propagation in environments that use passive metasurfaces designed to enhance wireless signal coverage by controlling how signals propagate through a scene. These surfaces have electromagnetic material properties that do not adhere to Snell's law, but instead redirect signals according to a gradient or diffuse them over a wider range of angles than a normal material surface.

The RO Scattering model provides a uniform ray description of electromagnetic wave scattering by locally periodic metasurfaces of polygonal shape. It is valid for metasurfaces whose bulk scattering coefficients are periodic functions of a phase parameter, which, in turn, is a continuous and smooth function of surface coordinates. The scattered field is expressed in terms of reflected, transmitted, and diffracted rays that do not generally obey conventional geometrical constraints (i.e., Snell's law and the Keller cone).

The integration of the RO Scattering model into Wireless InSite's X3D model enables ray-based EM propagation prediction in complex, electrically large environments, such as office buildings and urban areas equipped with metasurfaces.

### 2. Passive Metasurfaces Supported by EES Model

EES materials can be used to represent two types of passive metasurfaces for enhancing signal coverage in a scene: gratings and diffusers. Grating surfaces generally redirect incident signals in a prescribed direction, while diffusers spread the scattering rays over a wide range of angles. Diffusers and gratings can be one-dimensional or two-dimensional. Two dimensional EES's can have scattering characteristics that differ across the two principal dimensions of the


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metasurface, and can also redirect signals at an arbitrary polar angle away from the angle of incidence. These characteristics can include both reflective and transmissive properties, allowing placement on walls or windows, depending on the EES design parameters.

### 3. Control Interaction with EES Diffusers

Since an EES diffuser spreads ray paths in a range of directions, the simulation time can increase rapidly. Paths that interact multiple times with an EES diffuser are often low-powered and do not contribute appreciably to total received power. In order to reduce run-times, the user can specify An EES diffusers interaction limit. This does not affect interactions with non-EES materials.

## MIMO Improvements

### 4. MIMO Mutual Coupling via Multi-Port S-Parameters

Wireless InSite now supports the import of Touchstone S-parameter data to handle mutual coupling effects between MIMO antenna elements. This allows calculations to capture the resultant increase in correlation between signals transmitted and received by the antenna elements, affecting the ability to establish orthogonal streams and degrading overall performance of MIMO systems.

### 5. Receiver Codebooks

To improve realism for MIMO channels, users can now define codebooks for both transmitters and receivers. The model will then select the best matched beam pair for each point-to-point link for optimal performance when calculating results for points, routes, or coverage grids.

### 6. Closed-loop spatial multiplexing enhancements

This version provides increased control of spatial multiplexing, allowing the user to specify the maximum desired number of streams and choose from uniform or waterfilling power allocation schemes. The model will then ensure equal modulation across streams, and optionally select the optimal number of streams in terms of the resultant total peak throughput according to the user-specified power allocation method.

## Multi-Frequency Antennas

Antennas can now be defined in terms of full 3D patterns as a function of frequency, obtained from measurements or full-wave solvers. This includes direct import of multi-frequency antenna patterns generated by Remcom's XFdtd®. Frequency-specific patterns increase flexibility and accuracy by adjusting the pattern to the specific channel of a frequency band. They can also be used in conjunction with frequency sweeps to adjust the pattern to each frequency sample, greatly improving the accuracy when sweeping over a broad frequency band.


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## Maintenance Improvements

**7. Numerous fixes and enhancements to resolve bugs, improve usability and enhance performance**


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